Auteur: Etienne El Hachem
Studierichting: Informatica
Oefeningenreeks 6D nr. 30.12

Opgave: $\sum_{n=1}^{\infty} \frac{n \cdot \sin\left(\frac{1}{n}\right)}{\frac{1}{n} + 5}$

Criterium 1:

$$\lim_{n \to +\infty} \frac{n \cdot \sin\left(\frac{1}{n}\right)}{\frac{1}{n} + 5}$$

$$= \lim_{t \to 0} \frac{\frac{1}{t} \cdot \sin t}{t + 5}$$

$$= \lim_{t \to 0} \frac{\frac{\sin t}{t}}{t + 5} = \frac{1}{5}$$

$\Rightarrow$ Divergent

References